Forecasting the US Unemployment Rate

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Abstract

This article aims to improve the forecasting accuracy of the seasonally adjusted U.S. unemployment rate. Due to the non-stationarity of the data, two methods were employed: ARIMA and Time Series Decomposition model. The series was decomposed using the Hodrick-Prescott (HP) filter to separate the trend and cyclical components. While the trend was initially modeled using Fourier transforms to capture periodicity, poor fit led to the use of ARMA for the trend. Polynomial regression was applied to model the cyclical component. In-sample results showed that Time Series Decomposition provided a slightly better fit, but the ARIMA model proved more reliable for long-term forecasts.

1 Introduction

The U.S. unemployment rate, a critical economic indicator, exhibits both cyclical and trend-driven patterns. Accurate forecasting of this variable is essential for policymakers and economic analysts. In this paper, I explore various modeling approaches for predicting future unemployment rates, focusing on the ARIMA model, which is commonly used for unstationary time series, and time series decomposition methods like Hodrick–Prescott (HP) Filter. Additionally, I investigated polynomial regression for trend and structural breaks in the series using advanced statistical tests to refine our forecasting approach.

The paper is organized as follows. Section 2 presents the stationary analysis of the data. In Section 3, I build models using two different approaches and provide a detailed description of each. Section 4 focuses on the comparative analysis of

the two models, evaluating their relative performance. Finally, Section 5 reports the forecasting results.

2 Stationarity Analysis

Stationary models assume that the statistical properties of a process remain constant over time, particularly a fixed mean and a constant variance. Although we can visualize the time series data, as shown in Figure 1, to assess its stationarity, it is preferable to employ robust formal tests to determine stationarity more precisely.

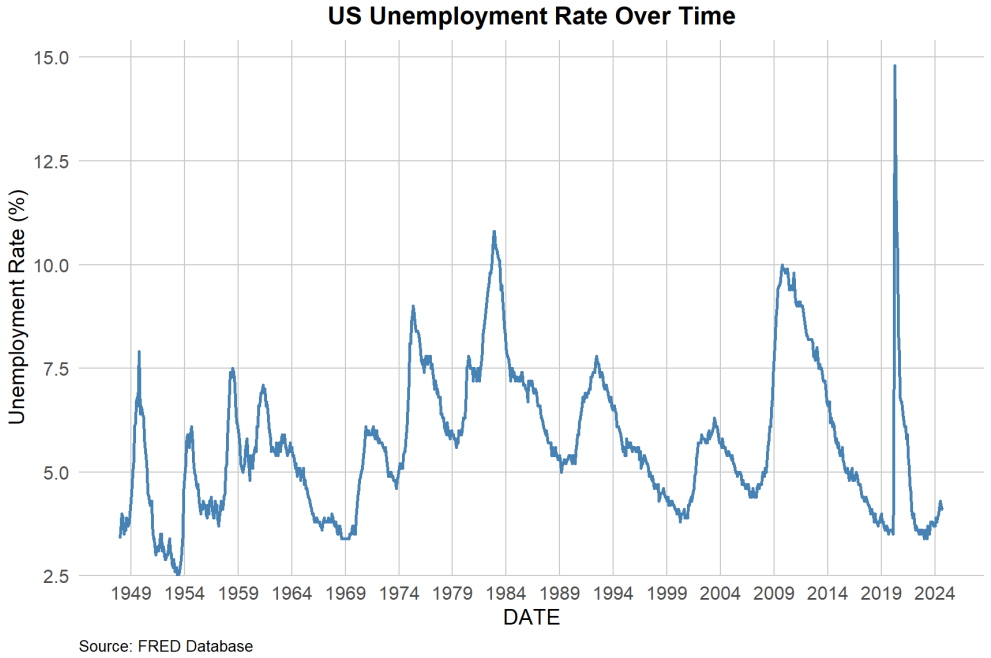


Figure 1: US Unemployment Rate Time Series Graph

2.1 Augmented Dickey–Fuller test

A commonly used statistical test for stationarity is the Augmented Dickey-Fuller (ADF) test (Dickey Fuller, 1979), which is a test for unit root.

Consider an AR(k+1) model:

$$\Delta y_t = \beta_0 + \phi y_{t-1} + \sum_{i=1}^{k} \gamma_i \Delta y_{t-i} + \epsilon_t \quad t = 1, 2, \ldots, \tag{1}$$

where $\Delta y_t = y_t - y_{t-1}$ represents the first difference of y_t, and ϵ_t is a white noise error term.

- If $|\phi| < 1$, the process is **stationary**, with values tending to revert to a mean over time, and its statistical properties (such as mean and variance) remain constant.
- If $|\phi| = 1$, the process exhibits a **unit root**, implying that the variance of the process increases over time ($\text{Var}(y_t) = t\sigma^2$), which indicates **non-stationarity**.
- If $|\phi| > 1$, the process becomes **explosive**, with values diverging over time.

After performing a regression analysis of historical U.S. unemployment rates against time, I observed that the t-value of the time coefficient was statistically significant (Table 1), leading to the inclusion of a time trend in the model:

$$\Delta y_t = \beta_0 + \beta_1 t + \phi y_{t-1} + \sum_{i=1}^{k} \gamma_i \Delta y_{t-i} + \epsilon_t \quad t = 1, 2, \ldots \tag{2}$$

The number of lag periods was determined using the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC). Both criteria selected a lag of 1. The PACF graph shows the same results (Figure 2). As shown in Table 2, the test statistic value is -3.8926, which is significantly smaller than the critical value of -3.41. Therefore, we conclude that the series is stationary under the regression model with a drift and trend term.

Variable	**Estimate**	**Std. Error**	**t value**	**Pr(>|t|)**
Intercept	5.489e+00	6.916e-02	79.369	$< 2 \times 10^{-16}$
Date	3.298e-05	6.868e-06	4.801	1.84e-06

Table 1: Regression Results for US Unemployment Rate against Date

Test Statistic	**Value**
ADF Test Statistic	-3.8926
Critical Values at 1%	-3.96
Critical Values at 5%	-3.41
Critical Values at 10%	-3.12

Table 2: ADF test results with a lag of 1 and a trend regression specification.

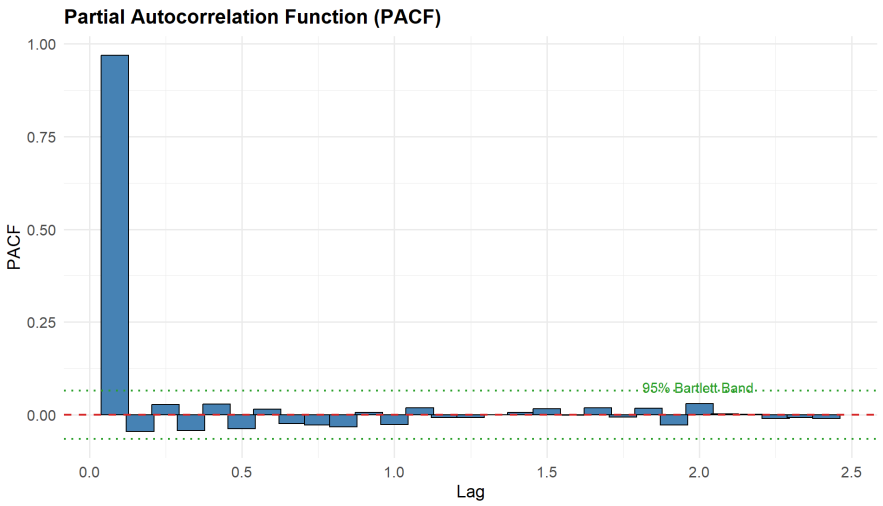


Figure 2: Partial Autocorrelation Function (PACF).

2.2 Kwiatkowski-Phillips-Schmidt-Shin (KPSS) Test

However, ADF test has its limitations, particularly its low power against the alternative hypothesis when the series is near-integrated or highly autoregressive (Diebold & Rudebusch, 1991; DeJong et al., 1992). Therefore, the KPSS test was employed in this analysis as a complementary approach.

After applying the KPSS test, I found that the value of the test statistic is 1.0112, which is significantly bigger than 0.739. This means we should reject the null hypothesis that the dataset is stationary, which is the opposite of the ADF test.

Test Statistic	Critical Value (10%)	Critical Value (5%)	Critical Value (1%)
1.0112	0.347	0.463	0.739

Table 3: KPSS Test Results

Although the seasonally adjusted unemployment rate removes seasonal fluctuations, it may not eliminate long-term trends or structural shifts. ADF test is not sensitive to weak trends, so it may be judged as stable; KPSS testing is very sensitive to trends, and even weak trends may be judged as non-stationary. Meanwhile, ADF test detects whether the data has a random trend, while KPSS test mainly detects the presence of deterministic trends. If the data has a certain linear or other deterministic trend (such as a long-term upward or downward trend over time), KPSS will consider the data non-stationary. Therefore, there is a high possibility of deterministic trends.

2.3 Zivot-Andrews Test

From the U.S. unemployment time series (Figure 3), a sharp deviation is observed around 2020, likely due to the economic impact of the pandemic. This suggests the presence of a structural break. Traditional tests, such as the ADF test, do not account for such breaks. Zivot and Andrews (1992) demonstrated that incorporating structural breaks can significantly alter unit root test results. Therefore, the Zivot-Andrews test was used to assess stationarity while allowing for an unknown structural break. Now the model is:

$$y_t = \alpha + \beta t + \phi y_{t-1} + \gamma_1 d_u + \gamma_2 d_t t + \epsilon_t \tag{3}$$

Where y_t is the time series value at time t, α is the intercept (constant), β is the coefficient on the time trend t, ϕ is the autoregressive parameter, d_u is a dummy variable representing the level break, d_t is a dummy variable representing the trend break, and ϵ_t is the error term.

After applying the Zivot-Andrews Test (Table 4), I found that the test statistic is -4.6254, which is bigger than -5.08. Therefore, we failed to reject the null hypothesis that the time series is nonstationary. Meanwhile, both p-values of the level break and trend break are smaller than 0.05, implying break points being included.

Instead of 2020 (Pandemic year) where we assume it will be the break point, Zivot-Andrews Test illustrates that 2008 is a potential break point. This timing coincides with the outbreak of the global financial crisis, which has had a profound impact on the United States and the global economy. The statistical characteristics of the unemployment rate may have changed around April 2008.

Test Statistic	**Critical Value (10%)**	**Critical Value (5%)**	**Critical Value (1%)**
-4.6254	-4.82	-5.08	-5.57

Table 4: Zivot-Andrews Test Results (1)

Variable	**Pr(>|t|)**
Level Break (d_u)	0.02531
Trend Break (d_t)	0.00511
Potential break point	2008 April

Table 5: Zivot-Andrews Test Results (2)

2.4 Hodrick–Prescott Filter

Based on the stationarity analysis, one possible explanation for the non-stationarity of U.S. unemployment is its countercyclical nature in economic theory. During periods of economic expansion, increased demand leads firms to expand their workforce, reducing unemployment. Conversely, in recessions, such as the 2008 financial crisis, reduced demand results in workforce cuts and higher unemployment. Additionally, exogenous shocks—like natural disasters and global pandemics—can disrupt labor markets, causing sharp, short-term increases in unemployment that deviate from typical business cycle patterns.

Given the possibility of deterministic trends (Section 2.2) and structural shifts (Section 2.3), I used the Hodrick-Prescott (HP) filter to decompose the time series into its trend and cyclical components. The HP filter extracts the long-term trend part from the data, allowing us to see the long-term trend of the unemployment rate more clearly without being disturbed by short-term fluctuations, such as economic cycle fluctuations (Hodrick & Prescott, 1997). The Hodrick-Prescott filter decomposition is:

$$y_t = \tau_t + c_t \tag{4}$$

where τ_t represents the long-term trend and c_t represents the cyclical component.

Figure 3 provides a clear depiction of the trend and cyclical fluctuations in the unemployment rate, offering valuable insights into its underlying dynamics.

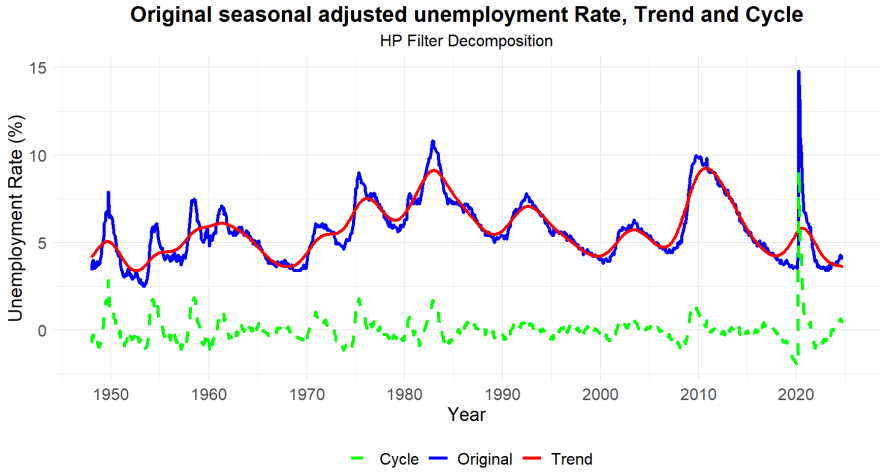


Figure 3: HP Filter

3 Forecasting Model

3.1 ARIMA Model

Since our time series is unstationary and ARMA Model can only be used for stationary time series data, ARIMA Model would be a good choice for forecasting. Consider an ARMA (p, q) Model,

$$y_t = \mu + \sum_{i=1}^{p} \phi_i y_{t-i} + \sum_{j=1}^{q} \theta_j \epsilon_{t-j} + \epsilon_t \tag{5}$$

Where y_t is the value of the time series at time t, μ is a constant (or trend term, if included), ϕ_i are the coefficients of the autoregressive (AR) part, θ_j are the coefficients of the moving average (MA) part, and ϵ_t is the white noise error term at time t, which is assumed to be a random disturbance with zero mean and constant variance.

It can be written as another form:

$$(1 - \phi_1 L - \phi_2 L^2 - \cdots - \phi_p L^p) y_t = (1 + \theta_1 L + \theta_2 L^2 + \cdots + \theta_q L^q) \epsilon_t \tag{6}$$

Where ϕ_i are the autoregressive coefficients, θ_j are the moving average coefficients, L is the lag operator, and ϵ_t is the error term.

The key point of ARIMA Model is applying finite differencing of the data, making it stationary. Consider an ARIMA (p, d, q) Model,

$$(1 - \phi_1 L - \phi_2 L^2 - \cdots - \phi_p L^p)(1 - L)^d y_t = (1 + \theta_1 L + \theta_2 L^2 + \cdots + \theta_q L^q) \epsilon_t \tag{7}$$

Where L is the lag operator, p, d, and q are the orders of the AR, differencing, and MA components, ϕ represents autoregressive coefficients, θ represents moving average coefficients, and ϵ_t is the error term.

As mentioned in Section 2.3, our data likely contains breakpoints. A single ARIMA model may not fully capture these changes, so segmenting the data and applying an appropriate ARIMA model for each segment is necessary. Additionally, since the unemployment rate is influenced by various cycles and shocks, the Zivot-Andrews test, which only allows for one structural break, may not be sufficient. Thus, I used the Bai-Perron test, which is more robust in detecting multiple breakpoints in the data (Bai & Perron, 1998).

As additional breakpoints improve model fit, they may also increase complexity and lead to overfitting. Since BIC penalizes complexity more than AIC, it is used

as the criterion for the Bai-Perron test.

m	0	1	2	3	4	5
BIC	3618	3483	3332	3290	3251	3257
Breakdates	m = 4	1974(10)	1987(3)	2001(10)	2013(4)	

Table 6: BIC values for different segments and Optimal segments' breakdates

Given that the BIC value for a model with four breakpoints is the smallest, the data were divided into five segments for the purpose of modeling with ARIMA. The results are illustrated in Figure 4.

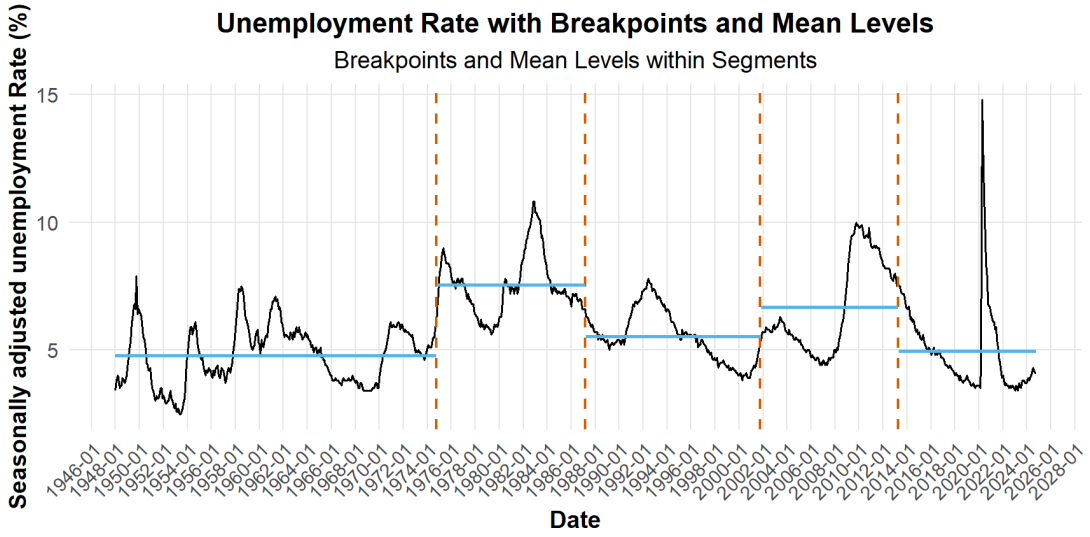


Figure 4: Mean shifts graph

We estimated an ARIMA model using data after the last breakpoint to forecast November's seasonally adjusted unemployment rate. The auto.arima function in R was employed to automatically select the optimal ARIMA (p, d, q) model. The point forecast for November's unemployment rate is 4.1%.

3.2 Time Series Decomposition

In Section 2.4, we have decomposed the time series into trend and cyclical parts, where the cyclical part is stationary. Therefore, we can use different models to estimate the two parts separately and add them up in the end to get the forecast value.

Figure 6 illustrates the strong periodicity in the cycle component, while its ACF reveals regular periodic fluctuations and values that surpass the Bartlett

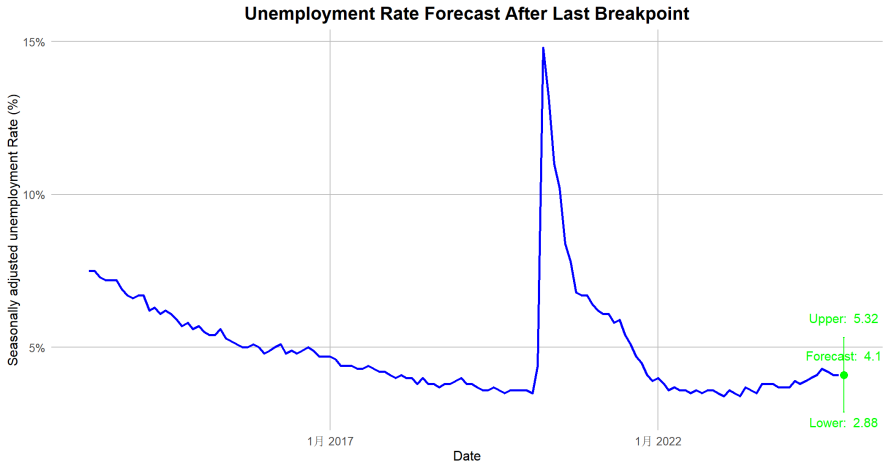


Figure 5: ARIMA Forecast of Unemployment Rate

confidence interval. Two models are considered for this analysis: Fourier analysis with a harmonic model and the classic ARMA model, commonly used for stationary time series data.

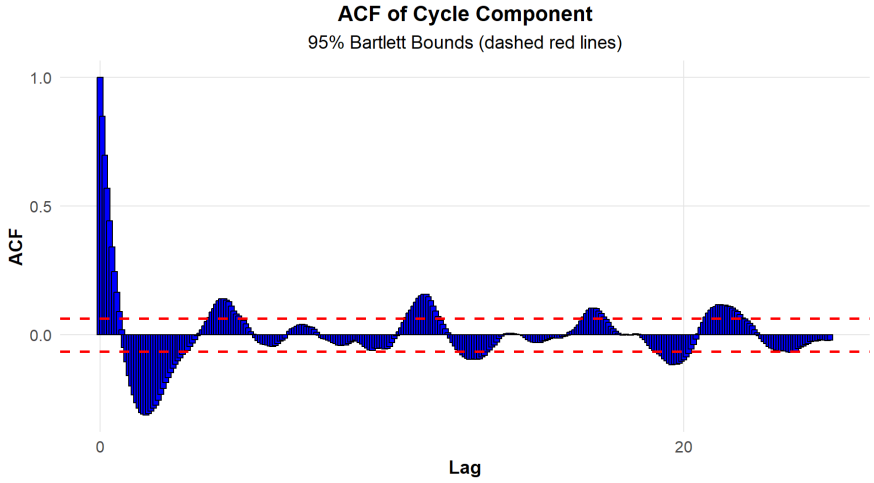


Figure 6: ACF of Cycle part

We first consider Fourier analysis and harmonic model. The key of Fourier analysis is that we assume that time series can be decomposed into sine and cosine components of varying frequencies (Fourier, 1822), revealing periodic patterns. A general Fourier series model is:

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos(n\omega t) + b_n \sin(n\omega t)) \tag{8}$$

Where a_0 is the average (DC) component, a_n and b_n are the Fourier coefficients,

ω is the fundamental frequency, and n is the harmonic number.

Assume we represent the cyclical time series data using a harmonic regression model:

$$y_t = \beta_0 + \beta_1 \cos(2\pi f t) + \beta_2 \sin(2\pi f t) + \epsilon_t \tag{9}$$

Where: β_0: Intercept term, β_1, β_2: Coefficients for the cosine and sine terms, f: Frequency of the cycle, and ϵ_t: Random error.

To identify the frequency components (cycles) of a time series, we apply Fourier transform. The Fourier Transform of a function *f(t)* is given by:

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t}\, dt \tag{10}$$

Where $F(\omega)$ represents the frequency-domain signal, ω is the angular frequency, and $e^{-i\omega t}$ is the complex exponential combining sine and cosine waves.

We can transform the time-domain signal into its frequency-domain representation using the Fast Fourier Transform (FFT) algorithm (Cooley & Tukey, 1965). The extracted frequency components were then applied in the harmonic regression model.

As discussed in Section 2.4, a potential breakpoint exists in the cyclical component. This breakpoint was incorporated for harmonic regression analysis. The results of this model are visualized in Figure 7.

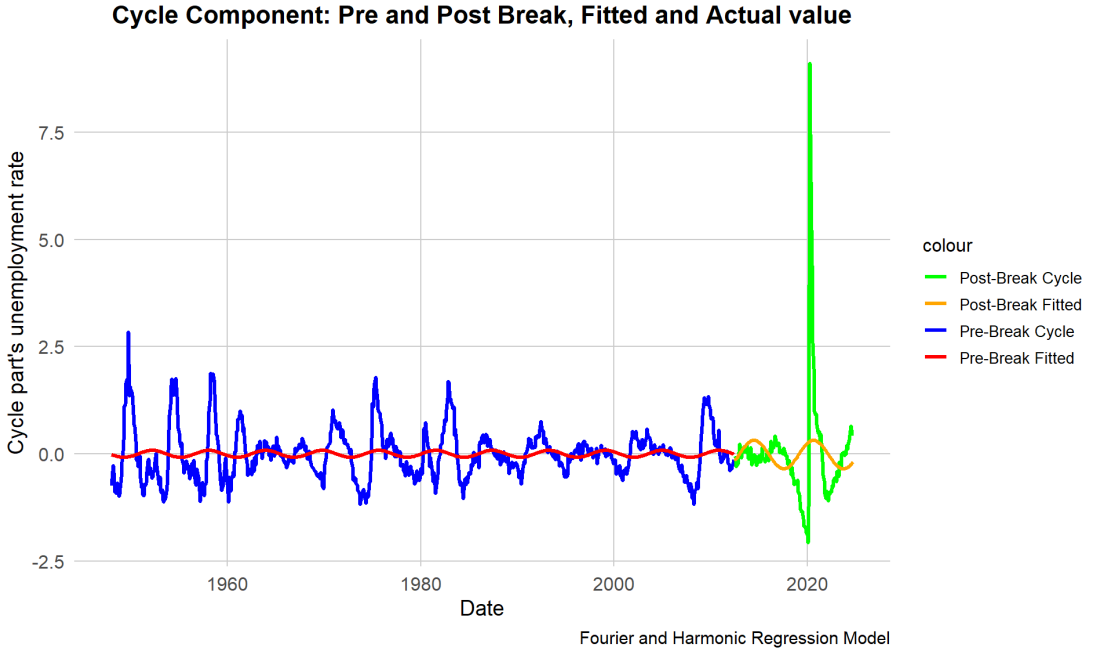


Figure 7: Harmonic regression

However, we can see that this model has poor fitting ability for periodic sequences from Figure 7. It has a very low R-square which is less than 0.2.

Next, we applied an ARMA model using R's auto function to select the best-fitting ARMA($p,\ q$) model, incorporating the same breakpoint as before. The R-squared value before the breakpoint is 0.9004, indicating a strong fit. However, after the breakpoint, it drops to 0.5768, likely due to the sharp changes during the epidemic, which are harder to model. Despite this, it still outperforms the Fourier and harmonic regression models. The forecast for the next month's cycle component is 0.35%.

Model	R-squared
Pre-break	0.9004
Post-break	0.5768

Table 7: R-squared values for ARMA models before and after the breakpoint

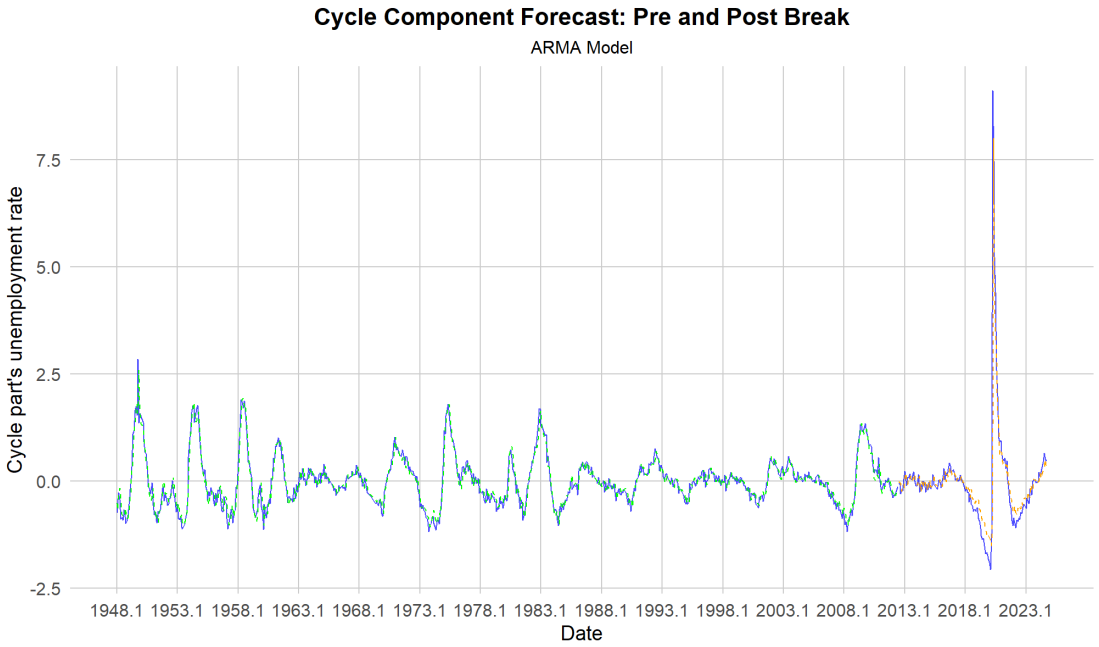


Figure 8: ARMA model for cycle

From Figure 9, we can see the trend part is really like a polynomial graph. Assume the trend part can be represented as:

$$y = \beta_0 + \beta_1 t + \beta_2 t^2 + \cdots + \beta_n t^n + \epsilon \tag{11}$$

Where y is the dependent variable (the value you want to predict), t is the independent variable (time or another feature), $\beta_0, \beta_1, \ldots, \beta_n$ are the coefficients

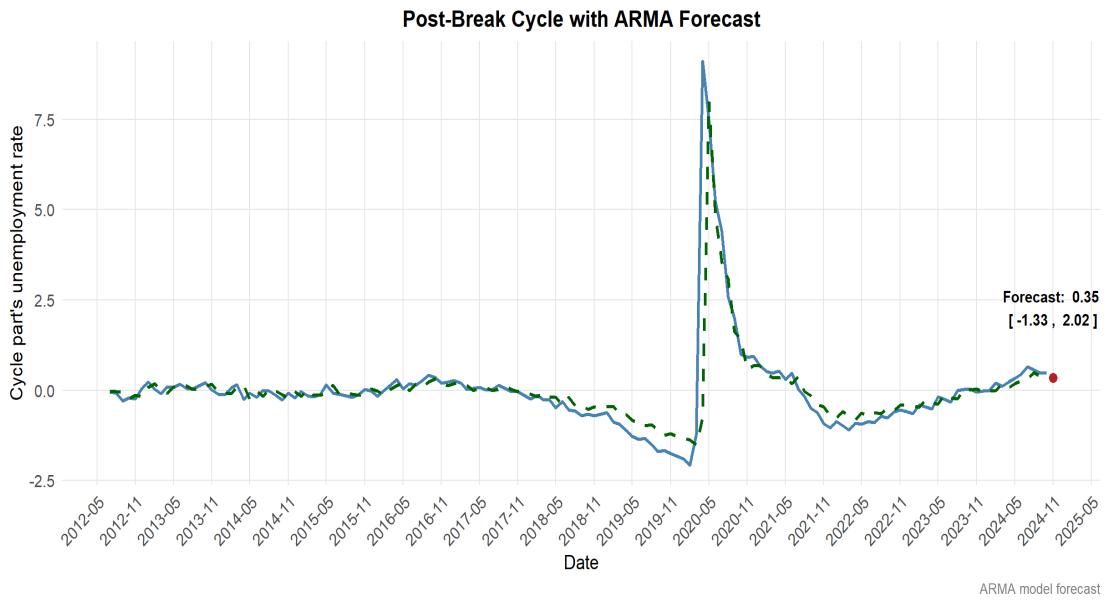


Figure 9: ARMA model forecast for cycle

that determine the shape of the polynomial, and ϵ is the error term, accounting for the difference between the predicted and actual values.

I applied the Bai-Perron test to detect breakpoints and then performed polynomial regression on each segment. For each segment,

4 Model Selection

4.1 Forecast Accuracy Evaluation

To evaluate the accuracy of the model's forecasts, I use several commonly applied measures of forecast accuracy: R-squared, RMSE (Root Mean Squared Error), MSPE (Mean Squared Prediction Error), MAE (Mean Absolute Error), and MAPE (Mean Absolute Percentage Error). R-squared indicates how well the model explains the variation in the observed data, reflecting its overall fit. However, it doesn't directly assess prediction accuracy. To address this, RMSE and MAE are used: RMSE penalizes large errors, highlighting models that minimize significant deviations, while MAE treats all errors equally, avoiding disproportionate penalties. MSPE emphasizes the importance of minimizing large prediction errors by focusing on squared differences between predicted and actual values. Lastly, MAPE normalizes errors as percentages, allowing for comparison of model accuracy across different datasets or scales. Together, it can provide a comprehensive evaluation of model performance.

Overall, ARIMA model performs well across all measures. The high R-squared (0.9427) indicates a good fit, while the low RMSE, MAE, MSPE, and MAPE values suggest accurate predictions with minimal errors. The low MAPE, in particular, shows that the model's prediction errors are small relative to the observed values, making it a strong candidate for forecasting.

The Time Series Decomposition model also demonstrates strong fitting ability. When comparing the two models, ARIMA performs better MAPE, indicating superior prediction accuracy. However, the Time Series Decomposition model performs slightly better in R-squared, RMSE, MSPE and MAE, suggesting it explains more variance and provides a better overall fit. In conclusion, while ARIMA is more accurate in predicting future values, the Time Series Decomposition model offers a better fit to the data.

Measures	**Value**
R-squared	0.9427
RMSE	0.4092
MSPE	0.1674
MAE	0.1576
MAPE	2.7674

Table 8: ARIMA Performance

Measures	**Value**
R-squared	0.9507
RMSE	0.3794
MSPE	0.1440
MAE	0.1542
MAPE	2.8317

Table 9: Decomposition Performance

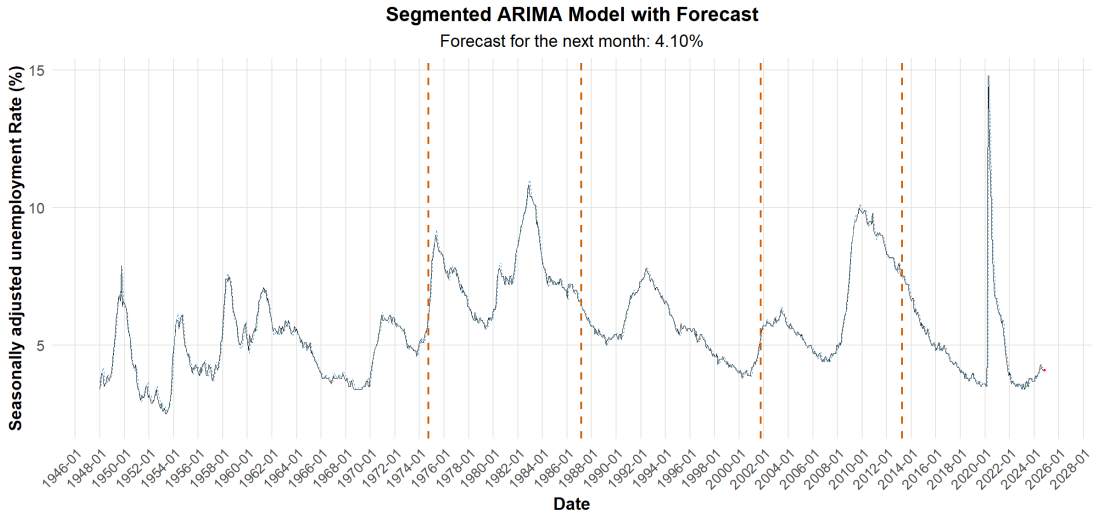


Figure 10: ARIMA Forecast of Unemployment Rate

4.2 Forecast Accuracy Tests

What we used in section 4.1 is in-sample evaluation. However, this approach can be overly optimistic, as fitting algorithms tend to minimize prediction errors, which may lead the model to capture random noise rather than meaningful patterns. This can result in overfitting, where the model is too closely tailored to the training data, making it less effective at forecasting future data. Due to the availability of only one dataset, real out-of-sample evaluation is not possible. Therefore, I will use pseudo out-of-sample forecast evaluation.

Pseudo out-of-Sample forecast evaluation is a method used to assess forecasting performance when only one dataset is available. Given a sample of $T + 1$ observations, the first R observations are used to estimate the initial model, and the subsequent observations are used for prediction. The last P observations of the sample serve as the out-of-sample period for forecast evaluation. The resulting forecasting errors $\{e_{t+l}\}_{l=R+1}^{T+1}$ are then used to evaluate the model's out-of-sample forecasting ability.

The evaluation procedure involves splitting the dataset into an in-sample estimation period and an out-of-sample forecasting period. The two competing models are estimated using the in-sample period, followed by the construction of forecasts and calculation of forecast errors during the out-of-sample period. Finally, statistical tests are conducted to compare the forecast accuracy of the two models.

I used three tests to compare forecast accuracy: the MSPE-based F test, Granger-Newbold test, Diebold-Mariano test. The MSPE-based F test compares the mean squared prediction errors (MSPE) of different models, with lower MSPE indicating better accuracy. The Granger-Newbold test checks if the model outperforms a naive benchmark. The Diebold-Mariano test assesses whether one model's forecast errors are significantly smaller than another's. Together, these tests offer a comprehensive evaluation of forecast performance.

Set time decomposition model as model 1 and ARIMA model as model 2. MSPE-based F test statistic is calculated as follows (Diebold Mariano, 1995):

$$F = \frac{\text{MSPE}_1}{\text{MSPE}_2} \sim F(N, N) \tag{12}$$

where MSPE_1 is the Mean Squared Prediction Error of Model 1, MSPE_2 is the Mean Squared Prediction Error of Model 2, and N is the number of sample size used for predictions.

The MSPE-based F test statistic of 0.8599 and its associated p-value of 0.9889 indicate that the null hypothesis cannot be rejected. This suggests there is no statistically significant difference between the predictive accuracy of the two models being compared.

Statistic	**Value**
MSPE-based F Test Statistic	0.8599
p-value for F Test	0.9889

Table 10: Results of MSPE-based F Test

The Granger-Newbold Test is defined as follows (Granger Newbold, 1974):

$$\text{GN} = \frac{r_{xz}}{\sqrt{\frac{1-r_{xz}^2}{N-1}}} \sim t(N-1), \tag{13}$$

where $x_i = e_{1i} + e_{2i}$, $z_i = e_{1i} - e_{2i}$, r_{xz} is the sample correlation coefficient between x_i and z_i, and N is the number of forecasts.

When r_{xz} is statistically different from zero, Model 1 has a larger MSPE if $r_{xz} > 0$, and Model 2 has a larger MSPE if $r_{xz} < 0$.

The Granger-Newbold test yields a p-value of 4.248253e-09, indicating strong evidence against the null hypothesis of no difference in forecast accuracy between the two models. The negative GN statistic suggests that Model 2 (ARIMA) has a larger MSPE, consistent with the findings in Section 4.1.

Test	**Value**
Granger-Newbold Test Statistic	-5.931314
p-value for Granger-Newbold Test	4.248253e-09

Table 11: Granger-Newbold Test Results

For one-step-ahead forecast error, define $d_i = g(e_{1i}) - g(e_{2i})$ and $\bar{d} = \frac{1}{N}\sum_{i=1}^{N} d_i$. The Diebold-Mariano Test is defined as follows (Diebold Mariano, 1995):

$$DM = \frac{\bar{d}}{\sqrt{\frac{\gamma_0 + 2\gamma_1 + \cdots + 2\gamma_q}{N-1}}} \sim t(N-1) \tag{14}$$

Where:γ_j is the j-th sample autocovariance of the d_i sequence,N is the number of forecasts.

The p-value from the Diebold-Mariano Test is between 5% and 10%. At a 95% confidence level, we cannot reject the null hypothesis, implying no significant difference in prediction accuracy between the two models. However, at a 90%

confidence level, we reject the null hypothesis, indicating a marginal difference in prediction accuracy between the models.

Test	Value
Diebold-Mariano Test Statistic	-1.7693
p-value for Diebold-Mariano Test	0.07685

Table 12: Diebold-Mariano Test Results

Overall, both models demonstrate a good fit in Section 4.2, but slight differences in forecast accuracy exist. Given that the Time Series Decomposition model performs better in forecast accuracy evaluation, I selected 4.01% as the predicted value for the seasonally adjusted US unemployment rate in November.

4.3 Long Term Prediction

The Time Series Decomposition model encounters a significant issue when forecasting over long periods, such as predicting monthly unemployment rates for the coming years. Due to the sharp spike in the trend component during the pandemic, polynomial regression tends to predict an unrealistic sharp increase in the future. As shown in Figure 12, for slightly more distant predictions, the model provides highly unrealistic values.

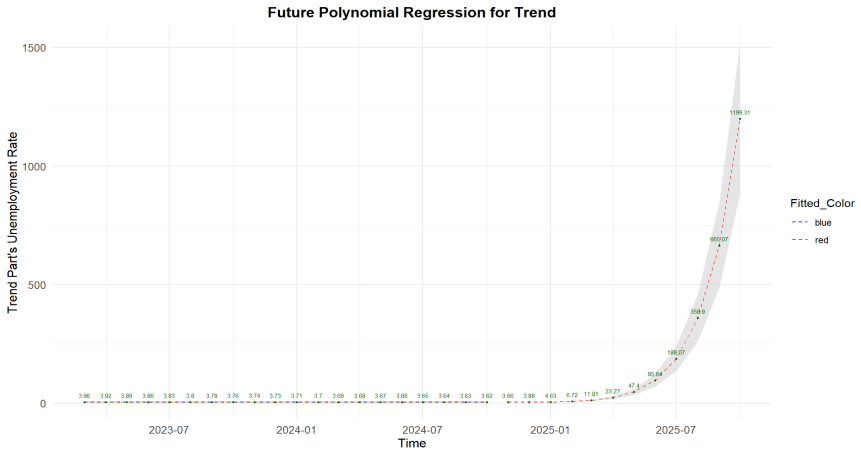


Figure 11: Longe term polynomial prediction

5 Results

Given that the Time Series Decomposition model is better suited for short-term forecasts, I have opted for the ARIMA model to predict the next twelve months.

The short-term seasonally adjusted US unemployment forecast for the next month is 4.01% based on the Time Series Decomposition model. For the longer-term forecast, the ARIMA model is used to predict the next twelve months, with results presented in Table 21.

Date	Forecast	95% Lower	95%Upper
2024-11	4.1027%	3.2869%	4.9185%
2024-12	4.1006%	2.9142%	5.2870%
2025-01	4.1022%	2.6566%	5.5479%
2025-02	4.1010%	2.4221%	5.7798%
2025-03	4.1020%	2.2282%	5.9757%
2025-04	4.1012%	2.0442%	6.1582%
2025-05	4.1018%	1.8814%	6.3221%
2025-06	4.1013%	1.7253%	6.4773%
2025-07	4.1017%	1.5822%	6.6212%
2025-08	4.1014%	1.4442%	6.7586%
2025-09	4.1016%	1.3150%	6.8882%
2025-10	4.1014%	1.1901%	7.0128%

Table 13: Forecast for the next 12 months of the seasonally adjusted unemployment rate.

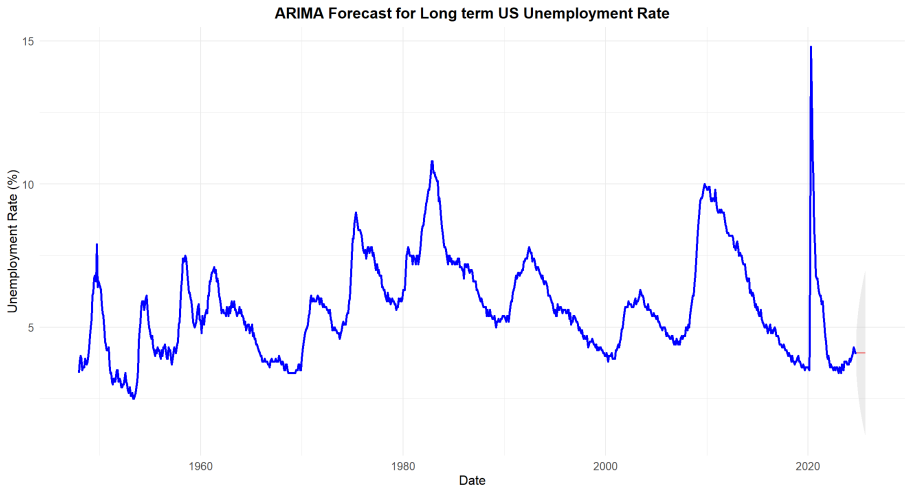


Figure 12: Long-term ARIMA prediction

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